TEST PLAN

Traffic Management System (Simulation)
Software Engineering Mini-Project

Document purpose: This document defines the test approach and contains the detailed test case records for the Traffic Management System (Simulation). The detailed cases are included within this Test Plan, not as a separate appendix.

Requirements baseline: Project SRS requirements FR1–FR8, NFR1–NFR3, SO1–SO2 and SR1–SR2, together with the project SADS.

Test case count: 12 (TC-01 to TC-12). Actual Result and execution status must be completed only after running each test against the final submitted build.

1. Introduction

1.1 Purpose

This Test Plan describes the scope, test items, features to be tested, test approach, security validation, execution process, test cases and traceability for the Traffic Management System (Simulation). It is intended to verify the implemented system against the approved project requirements.

1.2 Scope

Testing covers vehicle addition, per-lane FIFO queue handling, traffic signal states and transitions, vehicle movement, traffic information display, congestion classification, statistics, input validation, empty-queue protection, data integrity and representative workload behavior. A test may be marked Pass only when the final application demonstrates the expected result.

1.3 References

• Traffic Management System Software Requirements Specification (SRS).
• Traffic Management System Software Architecture and Design Specification (SADS).
• Software Engineering mini-project evaluation checklist and instructor-provided test-case format.

1.4 Definitions

FR: Functional Requirement; NFR: Non-Functional Requirement; SO: Security Objective; SR: Security Requirement; FIFO: First-In, First-Out; SRS: Software Requirements Specification; SADS: Software Architecture and Design Specification.

2. Test Strategy and Design Techniques

2.1 Test levels

Unit testing checks individual components such as vehicle and queue handling. Integration testing checks interactions among vehicle handling, queues, signal control and congestion detection. System testing checks end-to-end behavior against the SRS. Security validation checks input rejection and protection of queue state.

2.2 Test design techniques

Positive testing is used for valid operations; negative testing is used for invalid inputs and unsafe operations; boundary-value analysis is used for congestion thresholds; state-transition testing is used for traffic signals; and workload testing is used for representative input volumes.

2.3 Test environment and tools

Record the operating system, compiler and version, build command, test executable, test data and execution date for each run. The project test and coverage workflow uses C++ test programs and GCC coverage tooling such as gcov. Use the exact tool versions and commands recorded for the final execution.

3. Test Items

Test item	Description	Requirement coverage
Vehicle handling	Accept valid vehicle input and manage vehicle entry/removal.	FR1, FR5, SR1
Lane queues	Maintain per-lane FIFO order and safely handle queue operations.	FR2, FR5, NFR3, SR2
Signal controller	Maintain and transition supported traffic signal states.	FR3, FR4
Traffic display	Display current lane, signal and queue information.	FR6, NFR2

Test item	Description	Requirement coverage
Congestion detector	Classify congestion using the implemented per-lane thresholds.	FR7
Statistics	Calculate and display statistics supported by the application.	FR8
Input/state protection	Reject invalid input and prevent unsafe queue-state changes.	SO1, SO2, SR1, SR2

4. Features to Be Tested

Feature	Verification focus	Test cases
Add vehicle	Valid input is accepted; invalid input is safely rejected.	TC-01, TC-11
Queue order/removal	FIFO order is preserved; valid removal removes the front vehicle only.	TC-02, TC-10, TC-12
Signal/movement	Signal transitions follow the implemented cycle; vehicle movement follows signal rules.	TC-03, TC-04
Traffic display/statistics	Displayed information agrees with the observed application state.	TC-05, TC-08
Congestion	Boundary and below-threshold classifications match the implemented bands.	TC-06, TC-07
Workload/reliability	Representative workload completes; unsafe operations do not corrupt state.	TC-09, TC-10, TC-12

5. Test Approach and Validation

5.1 Security Validation

Security validation checks SO1 (Data Integrity), SO2 (Input Validation), SR1 (Input Validation) and SR2 (Queue Data Protection). It verifies that invalid lane or vehicle input is handled safely and that invalid queue operations do not corrupt stored vehicle or queue state.

Security objective/requirement	Validation method	Test case	Expected result
SO2 / SR1 — Input validation	Submit an invalid lane identifier and malformed vehicle input where supported.	TC-11	Invalid input is rejected or safely handled; no unintended queue modification occurs.
SR2 — Queue data protection	Attempt removal from an empty lane and inspect later queue operations.	TC-10	Empty-queue operation is safely handled; queue remains consistent.
SO1 — Data integrity	Perform valid and invalid operations and compare the final state with the expected state.	TC-12	Valid operations take effect; rejected operations cause no unintended state change.

5.2 Test Execution and Defect Handling

Execute each test using its recorded steps. Record the test environment, inputs, observed output, Actual Result, status and evidence. A test passes only when the observed result matches the Expected Result. For a failure, record the reproduction steps, expected versus actual behavior and supporting evidence; report the defect, correct it and retest. Do not pre-fill Actual Result or Pass/Fail before execution.

5.3 Entry and Exit Criteria

Entry: Final source code is available; the application builds; test environment and test data are prepared; and the relevant requirement is understood.
Exit: All planned tests have execution records; failures and retests are documented; coverage and test reports are generated from the final tested code; and unresolved issues are disclosed.

5.4 Pass/Fail Criteria

Pass: Actual behavior matches the Expected Result and no unintended state change occurs. **Fail:** Actual behavior differs, the application crashes or hangs, or state is corrupted. **Not Run:** The test has not been executed. **Blocked:** Execution cannot proceed because of an identified dependency or environment issue.

6. Detailed Test Cases

Each test case uses the instructor's test-case parameters. The additional Related SRS Requirements field makes traceability explicit within each individual case. Actual Result and Comments are to be completed from genuine execution evidence.

TC-01 — Add vehicle using valid input

Test Case ID	TC-01
Test Scenario	Add vehicle using valid input
Test Case Description	Verify that a vehicle can be added to a valid configured lane using a valid vehicle identifier.
Test Setup	Start the simulation and identify a configured lane.
Test Steps	1. Select a valid lane. 2. Enter a valid vehicle identifier. 3. Submit the add-vehicle operation. 4. Inspect confirmation and lane queue/count.
Prerequisite (if any)	The application is running and the selected lane is initialized.
Assumption	The lane and vehicle identifier follow the formats accepted by the application.
Test Parameters	Valid configured lane; valid vehicle ID (e.g., V1).
Expected Result	The vehicle is accepted and added to the selected lane; confirmation is displayed and queue count increases by one.
Related SRS Requirements	FR1, SR1
Actual Result	
Comments	Positive test; attach console screenshot or test-run evidence.

TC-02 — Verify FIFO queue order and removal

Test Case ID	TC-02
Test Scenario	Verify FIFO queue order and removal
Test Case Description	Verify that vehicles are maintained in first-in, first-out order and valid removal removes the front vehicle only.
Test Setup	Start the simulation and select an empty valid lane.
Test Steps	1. Add V1, V2 and V3 to the same lane in that order. 2. Check queue count. 3. Perform one valid movement/removal. 4. Inspect remaining queue and count.
Prerequisite (if any)	The lane exists and accepts vehicles; queue contents can be observed.
Assumption	The application uses FIFO queue behavior for each lane.
Test Parameters	Insert V1, V2, V3; expected first removal is V1.
Expected Result	Queue count is three before removal; V1 is removed first; V2 and V3 remain in order; count becomes two.
Related SRS Requirements	FR2, FR5, NFR3, SR2
Actual Result	
Comments	Positive queue-integrity test; attach output or unit-test evidence.

TC-03 — Verify traffic signal state transitions

Test Case ID	TC-03
Test Scenario	Verify traffic signal state transitions
Test Case Description	Verify that the traffic signal changes only among supported Red, Yellow and Green states according to the implemented cycle.
Test Setup	Start the simulation with signal states initialized.
Test Steps	1. Record initial signal state. 2. Advance the simulation using its normal update/tick mechanism. 3. Record state after each update. 4. Continue until a complete cycle is observed.
Prerequisite (if any)	Signal controller is initialized and the application can advance the signal cycle.
Assumption	The cycle order and timing are those implemented in the submitted application.
Test Parameters	Repeated signal updates; observed state sequence.
Expected Result	Only valid states are displayed and transitions follow the cycle implemented in the final source code.
Related SRS Requirements	FR3, FR4
Actual Result	
Comments	Positive state-transition test; attach console output or test evidence.

TC-04 — Verify vehicle movement against signal state

Test Case ID	TC-04
Test Scenario	Verify vehicle movement against signal state
Test Case Description	Verify that vehicle movement follows the traffic signal rules implemented by the simulation.
Test Setup	Start the simulation and prepare a valid lane containing at least one vehicle.
Test Steps	1. Record lane signal and queue. 2. Observe movement while movement is not permitted. 3. Advance the normal signal cycle. 4. Observe movement when permitted. 5. Check queue count.
Prerequisite (if any)	A valid lane contains at least one vehicle and the signal controller is active.
Assumption	Movement follows the signal rules implemented in the submitted application.
Test Parameters	Vehicle ID; observed signal states; queue count before and after movement.
Expected Result	A vehicle moves only when the implemented signal rule permits movement; queue state updates accordingly.
Related SRS Requirements	FR3, FR4, FR5
Actual Result	
Comments	Integration test across signal, vehicle and queue handling.

TC-05 — Check traffic information displayed to the user

Test Case ID	TC-05
Test Scenario	Check traffic information displayed to the user
Test Case Description	Verify that displayed traffic information represents the current lane, signal and queue state.
Test Setup	Start the simulation and prepare a known lane, signal state and queue.
Test Steps	1. Add a known number of vehicles. 2. Note signal state and queue count. 3. Request/display traffic information. 4. Compare displayed values with known state.
Prerequisite (if any)	At least one configured lane has a known signal state and queue count.
Assumption	The display reports current application state without changing it.
Test Parameters	Selected lane; known queue count; observed signal state.
Expected Result	The output displays traffic information consistent with the known simulation state.
Related SRS Requirements	FR6, NFR2
Actual Result	
Comments	Positive display/usability test; attach console screenshot.

TC-06 — Check congestion at classification boundaries

Test Case ID	TC-06
Test Scenario	Check congestion at classification boundaries
Test Case Description	Verify congestion classification at the boundary values of the implemented per-lane congestion bands.
Test Setup	Start the simulation and select a valid lane for which queue length and congestion status can be observed.
Test Steps	1. Prepare a lane with 2 vehicles and record status. 2. Prepare it with 3 vehicles and record status. 3. Prepare it with 5 vehicles and record status. 4. Prepare it with 6 vehicles and record status.
Prerequisite (if any)	The congestion indicator is available and the selected lane can be populated to the required counts.
Assumption	The submitted implementation uses LOW for 0–2, MEDIUM for 3–5 and HIGH for 6 or more vehicles.
Test Parameters	Queue length 2 = LOW; 3 = MEDIUM; 5 = MEDIUM; 6 = HIGH.
Expected Result	The application reports LOW at 2, MEDIUM at 3 and 5, and HIGH at 6 vehicles.
Related SRS Requirements	FR7
Actual Result	
Comments	Boundary-value test. Attach output showing each queue length and its reported classification.

TC-07 — Verify low congestion classification

Test Case ID	TC-07
Test Scenario	Verify low congestion classification
Test Case Description	Verify that a lane containing two vehicles is classified as LOW congestion.
Test Setup	Start the simulation and select a valid lane with congestion status available.
Test Steps	1. Select a valid lane. 2. Add exactly 2 vehicles. 3. Request or observe congestion status. 4. Record the result.
Prerequisite (if any)	The lane exists and the congestion indicator is available.
Assumption	The implemented LOW band is 0–2 vehicles.
Test Parameters	Queue length = 2 vehicles.
Expected Result	The application reports LOW congestion for the selected lane.
Related SRS Requirements	FR7
Actual Result	
Comments	Below/at-boundary test; attach console output.

TC-08 — Verify traffic statistics

Test Case ID	TC-08
Test Scenario	Verify traffic statistics
Test Case Description	Verify that statistics reflect the vehicle operations performed during the test.
Test Setup	Start the simulation and record initial statistics/state.
Test Steps	1. Add a known set of vehicles. 2. Perform a known number of valid movements/removals. 3. Request statistics. 4. Compare reported values with observed operations.
Prerequisite (if any)	Statistics functionality is available and operations can be counted.
Assumption	Only statistics actually defined and displayed by the application are evaluated.
Test Parameters	Number of vehicles added and moved; displayed statistics.
Expected Result	Reported statistics match observed operations and calculation rules implemented by the application.
Related SRS Requirements	FR8
Actual Result	
Comments	Do not claim a metric that is not implemented by the application.

TC-09 — Exercise a representative high workload

Test Case ID	TC-09
Test Scenario	Exercise a representative high workload
Test Case Description	Check whether the application processes a larger controlled workload without crashing or becoming unusable.
Test Setup	Build and start the application in a clean state; record the test environment.
Test Steps	1. Choose and record a workload size. 2. Add vehicles across valid lanes. 3. Run signal updates and supported operations. 4. Observe completion and application behavior.
Prerequisite (if any)	The application starts and accepts normal vehicle operations.
Assumption	The workload is representative for this project and machine; no numerical response-time target is assumed unless defined in the SRS.
Test Parameters	Workload size; machine/compiler details; observed execution time if measured.
Expected Result	The workload completes without crash, hang or freeze; record timing without inventing an acceptance threshold.
Related SRS Requirements	NFR1
Actual Result	
Comments	Controlled workload test; attach output and environment details.

TC-10 — Protect an empty queue from invalid removal

Test Case ID	TC-10
Test Scenario	Protect an empty queue from invalid removal
Test Case Description	Verify that removing from an empty lane is safely handled and does not corrupt later queue operations.
Test Setup	Start the simulation and select a valid lane with an empty queue.
Test Steps	1. Attempt removal/dequeue from the empty lane. 2. Observe response. 3. Add V1 and V2. 4. Perform valid removals. 5. Verify remaining queue.
Prerequisite (if any)	The selected lane exists and is empty before the first operation.
Assumption	The queue component/application guards empty-queue removal.
Test Parameters	Removal from empty queue; then enqueue V1 and V2 and remove validly.
Expected Result	The empty-queue operation is safely handled; no crash or state corruption occurs; later valid operations preserve FIFO order.
Related SRS Requirements	FR2, FR5, NFR3, SR2
Actual Result	
Comments	Negative reliability and queue-protection test; attach test output.

TC-11 — Reject invalid lane or vehicle identifiers

Test Case ID	TC-11
Test Scenario	Reject invalid lane or vehicle identifiers
Test Case Description	Verify that invalid input is rejected before it can create a vehicle or modify a queue.
Test Setup	Start the simulation and identify valid lane identifiers and accepted vehicle-ID format.
Test Steps	1. Submit add-vehicle request with a lane ID that does not exist. 2. Observe response and queue state. 3. If supported, repeat with malformed vehicle ID. 4. Confirm no invalid vehicle was added.
Prerequisite (if any)	The simulation is running and valid identifier rules are known.
Assumption	Invalid values are selected according to validation rules in the submitted application.
Test Parameters	One invalid lane ID; optionally malformed vehicle ID if the interface supports it.
Expected Result	Invalid input is rejected or safely handled; no unintended vehicle or queue modification occurs.
Related SRS Requirements	FR1, SO2, SR1
Actual Result	
Comments	Negative/security validation test; attach console output.

TC-12 — Check data integrity after mixed operations

Test Case ID	TC-12
Test Scenario	Check data integrity after mixed operations
Test Case Description	Verify that valid operations update the simulation correctly and rejected operations do not corrupt vehicle, queue or signal state.
Test Setup	Start the simulation and prepare a lane with known initial queue and signal state.
Test Steps	1. Record initial queue, signal and available statistics. 2. Perform valid add/movement operations. 3. Attempt invalid input and empty-queue removal where supported. 4. Inspect final state. 5. Compare with expected state from valid operations only.
Prerequisite (if any)	The initial state can be observed and relevant operations are available.
Assumption	Rejected invalid operations should not change application state; valid operations should have their documented effects.
Test Parameters	Known initial queue; valid vehicle operations; invalid identifier; empty-queue removal where supported.
Expected Result	Valid operations are reflected correctly; rejected operations have no unintended effect; no vehicle is lost, duplicated or placed in an invalid queue.
Related SRS Requirements	FR1, FR2, FR5, SO1, SR2, NFR3
Actual Result	
Comments	Combined integration and security/data-integrity test; attach output.

7. SRS Traceability Matrix

This matrix provides overall traceability between the SRS requirements and the test cases. Requirement IDs are also recorded inside each individual test case above.

Requirement ID	Requirement area	Test case(s)
FR1	Add vehicle	TC-01, TC-11, TC-12
FR2	Maintain per-lane queues	TC-02, TC-10, TC-12
FR3	Signal transitions	TC-03, TC-04
FR4	Signal states	TC-03, TC-04
FR5	Vehicle entry/exit	TC-02, TC-04, TC-10, TC-12

Requirement ID	Requirement area	Test case(s)
FR6	Display traffic information	TC-05
FR7	Detect congestion	TC-06, TC-07
FR8	Calculate/display statistics	TC-08
NFR1	Performance/workload	TC-09
NFR2	Usability/readability of output	TC-05
NFR3	Reliability/queue state	TC-02, TC-10, TC-12
SO1	Data integrity	TC-12
SO2	Input validation	TC-11
SR1	Input validation	TC-01, TC-11
SR2	Queue data protection	TC-02, TC-10, TC-12

8. Test Organization, Entry/Exit Criteria and Deliverables

Testing is performed by the project team. The team member executing a test records the environment, test data, actual result, status and evidence. Defects are reported to the developer responsible for the affected feature and retested after correction.

8.1 Test deliverables

- This Test Plan with detailed test-case records.
- Test execution evidence and defect records.
- Unit Testing Report and System Validation Report.
- Code Coverage Report and Branch Coverage Report.
- Updated test cases reflecting final sprint changes.

8.2 Final execution record

For every test case, complete Actual Result and Comments after execution. Mark Pass only when the observed result satisfies the Expected Result; otherwise mark Fail and record the defect/evidence reference. If a test has not been run, mark it Not Run rather than assuming success.

9. Assumptions, Constraints and Items Requiring Confirmation

1. Congestion bands in TC-06 and TC-07 are LOW = 0–2 vehicles, MEDIUM = 3–5 vehicles and HIGH = 6 or more vehicles, matching the per-lane congestion feature previously implemented. Confirm the final submitted source still uses these thresholds.
2. The available SRS baseline does not provide a numeric response-time target for NFR1. TC-09 therefore records workload and observed timing; do not claim a numeric performance pass criterion unless an approved SRS target exists.
3. Execute only operations supported by the final application. If a step cannot be performed because a feature is absent, record Blocked/Not Applicable with an explanation and reconcile the SRS and test plan before submission.
4. Use actual names and behavior from the final source code when demonstrating implementation.